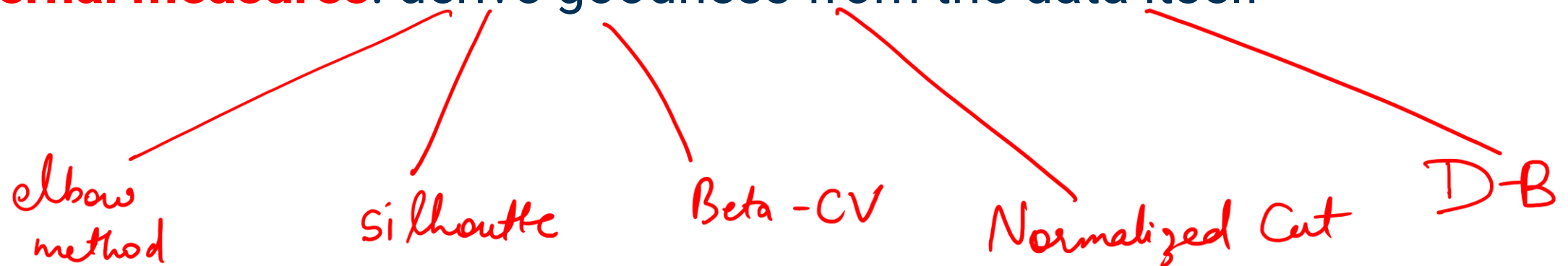


Clustering Evaluation

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Clustering Evaluation

- Clustering evaluation aims at quantifying the goodness or quality of the clustering.
- Two main categories of measures:
 - **External measures**: employ external ground-truth
 - **Internal measures**: derive goodness from the data itself



Outline

- External measures for clustering evaluation
 - Matching-based measures
 - Entropy-based measures
 - Pairwise measures
- Internal measures for clustering evaluation
 - Graph-based measures
 - Davies-Bouldin Index
 - Silhouette Coefficient

External Measures

External measures assume that the correct or ground-truth clustering is known *a priori*, which is used to evaluate a given clustering.

Let $\mathbf{D} = \{\mathbf{x}_i\}_{i=1}^n$ be a dataset consisting of n points in a d -dimensional space, partitioned into k clusters. Let $y_i \in \{1, 2, \dots, k\}$ denote the ground-truth cluster membership or label information for each point.

The ground-truth clustering is given as $\mathcal{T} = \{T_1, T_2, \dots, T_k\}$, where the cluster T_j consists of all the points with label j , i.e., $T_j = \{\mathbf{x}_i \in \mathbf{D} | y_i = j\}$. We refer to $\mathcal{T}$ as the ground-truth *partitioning*, and to each T_j as a *partition*.

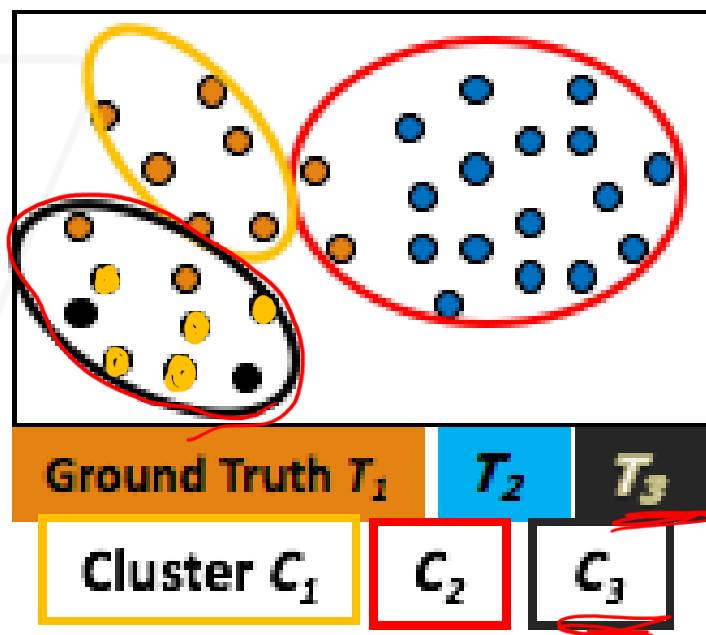
Let $\mathcal{C} = \{C_1, \dots, C_r\}$ denote a clustering of the same dataset into r clusters, obtained via some clustering algorithm, and let $\hat{y}_i \in \{1, 2, \dots, r\}$ denote the cluster label for $\mathbf{x}_i$.

So k is the number of ground truth partitions (T) and r is the number of clusters (C) obtained by algorithm

n_{ij} = Number of data points in cluster i which are also in ground truth partition j

Matching-Based Measures (I): Purity

- Purity**: Quantifies the extent that cluster C_i contains points only from one (ground truth) partition:



$$purity_i = \frac{1}{n_i} \max_{j=1}^k \{n_{ij}\}$$

$$purity_3 = \frac{1}{n_3} \max(n_{31}, n_{32}, n_{33}) = \frac{7}{9}$$

Handwritten notes for the calculation above:

- $i=3$ (pointing to the subscript 3 in n_3)
- $n_{31} = 2$, $n_{32} = 0$, $n_{33} = 7$ (values for the max function)
- $n_{31} = 7$, $n_{32} = 0$, $n_{33} = 2$ (values for the denominator n_3)
- $\frac{7}{9}$ (final result)

The Total purity of clustering C is the weighted sum of the cluster-wise purity:

$$purity = \sum_{i=1}^r \frac{n_i}{n} purity_i = \frac{1}{n} \sum_{i=1}^r \max_{j=1}^k \{n_{ij}\}$$

What is purity value for a perfect clustering?

Handwritten calculation for perfect clustering:

$$Purity = \frac{\sum n_i}{n} = \frac{n}{n} = 1$$

$$purity_i = \frac{1}{n_i} \max_{j=1}^k \{n_{ij}\}$$

$$purity = \sum_{i=1}^r \frac{n_i}{n} purity_i = \frac{1}{n} \sum_{i=1}^r \max_{j=1}^k \{n_{ij}\}$$

cluster $\nearrow$ n_{ij} $\longleftarrow$ *partition*

Example:

$$\text{purity}_1 = 30/50;$$

$$\text{purity}_2 = 20/25;$$

$$\text{purity}_3 = 25/25;$$

$$purity = (30 + 20 + 25)/100 = 0.75$$

$C \backslash T$	T_1	T_2	T_3	Sum
C_1	0	20	30	50
C_2	0	20	5	25
C_3	25	0	0	25
m_j	25	40	35	100

Two clusters may be matched to the same partition

C1 is more paired with T3
C2 is more paired with T2

C\T	T ₁	T ₂	T ₃	Sum
C ₁	0	20	30	50
C ₂	0	20	5	25
C ₃	25	0	0	25
m _j	25	40	35	100

$$\text{purity} = (30 + 20 + 25)/100 = 0.75$$

C1 is more paired with T2
C2 is more paired with T2

C\T	T ₁	T ₂	T ₃	Sum
C ₁	0	30	20	50
C ₂	0	20	5	25
C ₃	25	0	0	25
m _j	25	50	25	100

$$\text{purity} = (30 + 20 + 25)/100 = 0.75$$

Maximum weight matching: Only one cluster can match one partition

Ex. If C1 is more paired with T2 **THEN** C2 and C3 cannot paired with T2

C\T	T ₁	T ₂	T ₃	Sum
C ₁	0	30	20	50
C ₂	0	20	5	25
C ₃	25	0	0	25
m _j	25	50	25	100

MAX

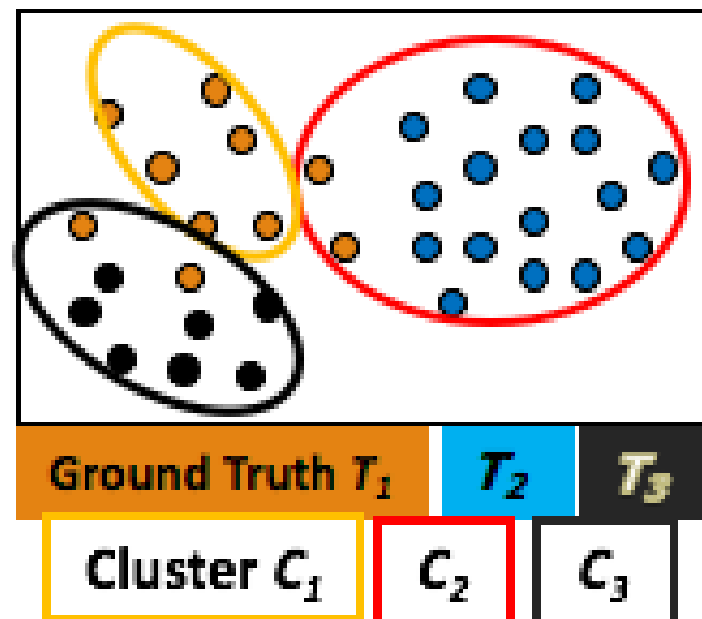
C1 is more paired with T2 = $\frac{30+5+25}{100} = 0.6$

C1 is more paired with T3 = $\frac{20+20+25}{100} = 0.65$

Purity = 0.65

Matching-Based Measures (II): Maximum Matching

- **Drawback of purity**: two clusters may be matched to the same partition.
- **Maximum matching**: the maximum purity under the one-to-one matching constraint.
 - Examine all possible pairwise matching between C and T and choose the best (the maximum)



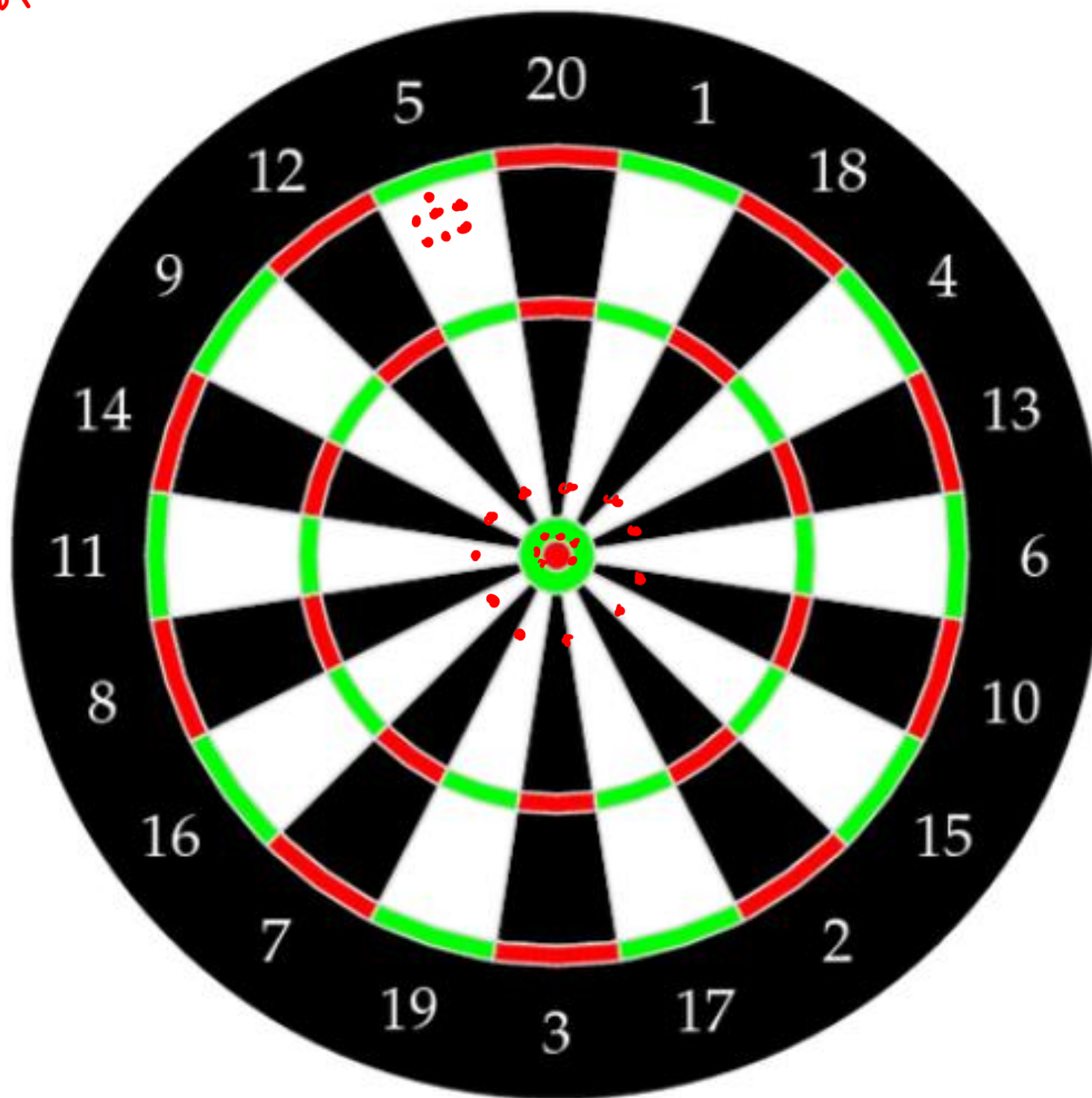
$C \backslash T$	T_1	T_2	T_3	Sum
C_1	0	30	20	50
C_2	0	20	5	25
C_3	25	0	0	25
m_j	25	50	25	100

Example:

Maximum matching = $0.65 > 0.6$

ML $\rightarrow$ my model is 95% precise $\times \rightarrow$ local solution

Accuracy $\rightarrow$ global measure
Precision



In a general context: Precision, Recall and Accuracy



positive → cat
negative → dog

Correct prediction

Wrong prediction

Number of predicted “positive” labeled data = True Positive + False Positive

Number of predicted “negative” labeled data = True Negative + False Negative

Precision = $\frac{\text{True Positive}}{\text{Predicted Results}}$ or

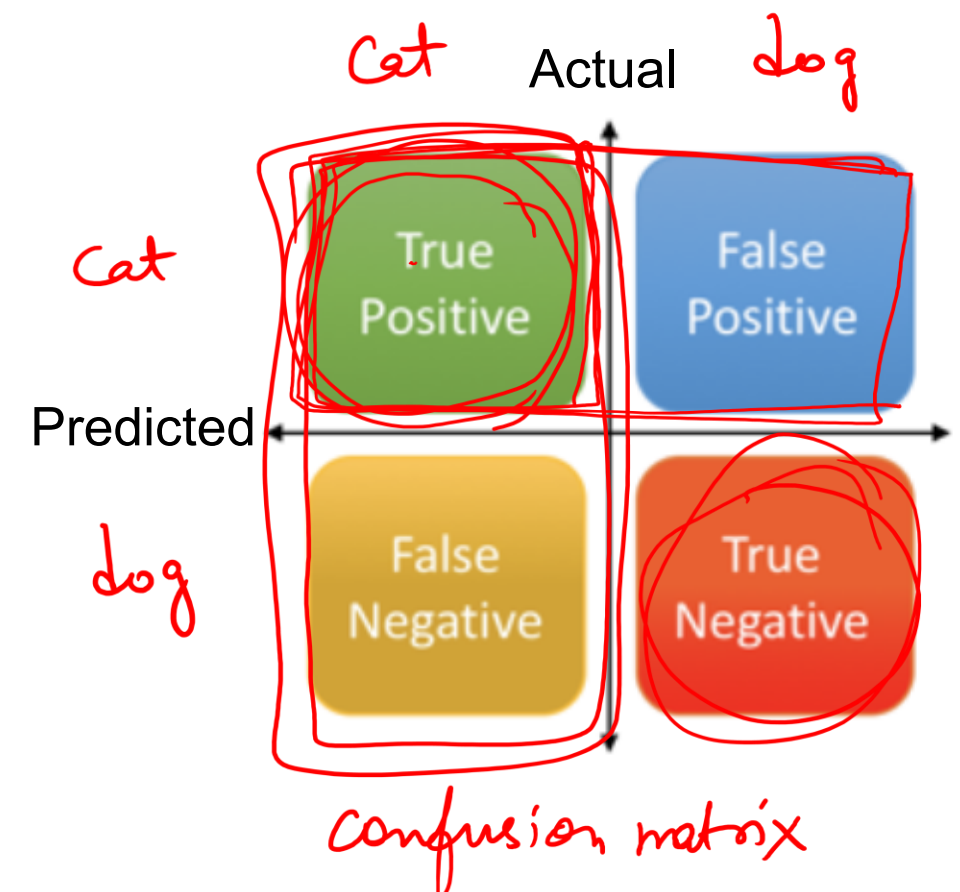
$\frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$

Recall = $\frac{\text{True Positive}}{\text{Actual Results}}$ or

$\frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$

Accuracy = *global*

$\frac{\text{True Positive} + \text{True Negative}}{\text{Total}}$



False positive is also called false alarm

Matching-Based Measures (II): F-Measure

- **Precision**: which measure *quality*, is the same as purity:

- How precisely does each cluster represent the ground truth?
- Of all predicted positives, how many are actually positive?

- **Recall**: measures completeness

- How completely does each cluster recover the ground truth?
- Of all actual positives, how many were correctly predicted?

$$recall_i = \frac{n_{ij_i}}{|T_{j_i}|} = \frac{n_{ij_i}}{m_{j_i}} \quad Recall_1 = \frac{6}{10}$$

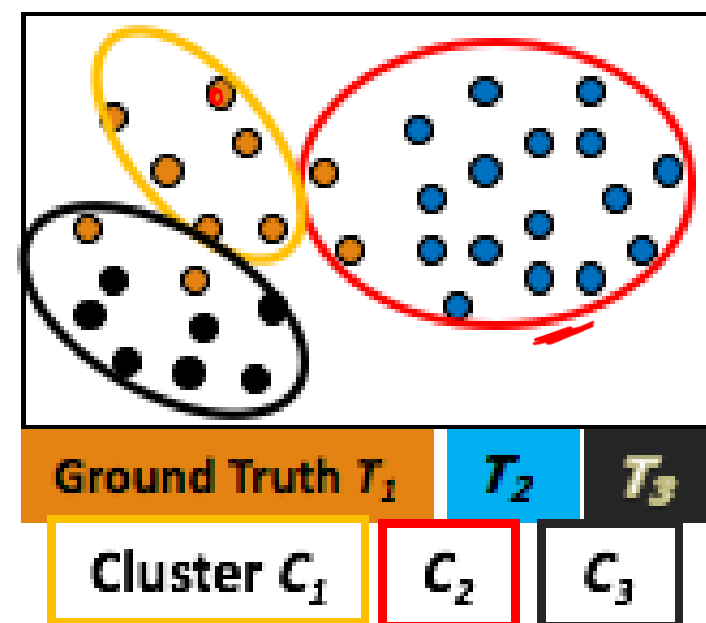
$$prec_i = \frac{1}{n_i} \max_{j=1}^k \{n_{ij}\} = \frac{n_{ij_i}}{n_i} \quad Prec_1 = \frac{6}{6} = 1$$

- C_i = i -th **cluster** produced by the algorithm

- T_j^* = **ground truth partition** that best matches cluster C_i

- n_{ij}^* = number of shared data points between C_i and T_j^*

- m_j^* = total number of points in that ground truth partition T_j^*



$$\frac{6}{10} \quad prec_2 = \frac{16}{18} \quad recall_2 = 1$$

Precision and Recall

(Precision here is same as the purity)

Precision:

$$\text{prec}_1 = 30/50;$$

$$\text{prec}_2 = 20/25;$$

$$\text{prec}_3 = 25/25$$

Recall:

$$\text{recall}_1 = 30/35;$$

$$\text{recall}_2 = 20/40;$$

$$\text{recall}_3 = 25/25$$

n_{ij}

$C \backslash T$	T_1	T_2	T_3	Sum
C_1	0	20	<u>30</u>	50
C_2	0	<u>20</u>	5	25
C_3	<u>25</u>	0	0	25
m_j	<u>25</u>	<u>40</u>	<u>35</u>	100

Matching-Based Measures (II): F-Measure

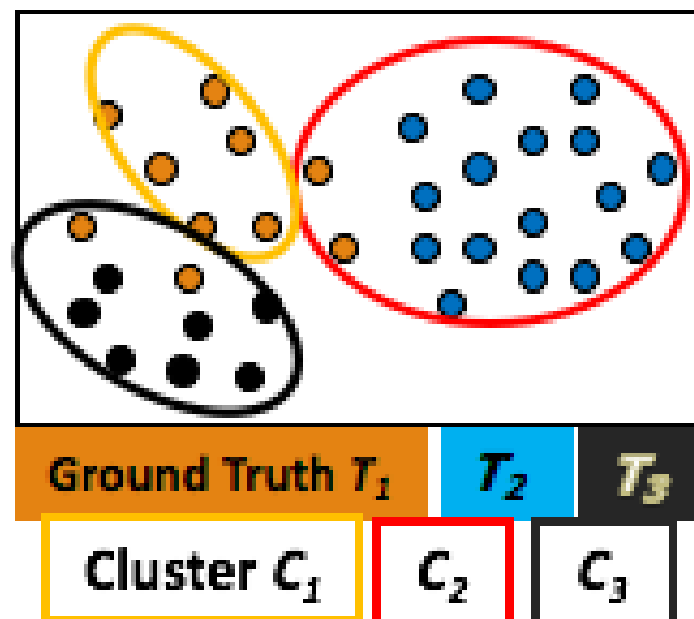
- **F-Measure**: the harmonic mean of precision and recall
 - Take into account both *precision* and *completeness*

$$F_i = \frac{2}{\frac{1}{prec_i} + \frac{1}{recall_i}} = \frac{2 \cdot prec_i \cdot recall_i}{prec_i + recall_i} = \frac{2 n_{ij_i}}{n_i + m_{j_i}}$$

The F-measure for the clustering $\mathcal{C}$ is the mean of clusterwise F-measure values:

$$F = \frac{1}{r} \sum_{i=1}^r F_i$$

$F_1 = 60/85;$
 $F_2 = 40/65;$
 $F_3 = 1;$
 $F = 0.774$



$C \backslash T$	T_1	T_2	T_3	Sum n_i
C_1	0	20 0	30	50
C_2	0	20	5	25
C_3	25	0	0	25
m_j	25	40	35	100

Entropy-Based Measures

- Expectation $E[g(x)] = \sum p(x) \cdot g(x)$
- Information $-\log_2 p(x)$
- Entropy $H(x) = \sum p(x) \cdot I(x) = -\sum p(x) \log_2 p(x)$
- We want entropy of a cluster to be high or low? $E(c_1) = 0$
- Conditional Entropy $H(Y|X_i) = -\sum p(Y|X_i) \log_2 p(Y|X_i)$
 $H(Y|X) = \sum_x p(x) \cdot H(Y|x_i)$

Entropy-Based Measures (I): Conditional Entropy

Amount of information orderliness in different partitions

- The entropy for clustering C and partition T is

$$\underline{H(C)} = - \sum_{i=1}^r p_{C_i} \log p_{C_i} \quad \underline{H(T)} = - \sum_{j=1}^k p_{T_j} \log p_{T_j}$$

where $\underline{p_{C_i}} = \frac{n_i}{n}$ and $\underline{p_{T_j}} = \frac{m_j}{n}$
i.e., The probability of ground truth T_j

$$\begin{aligned} p(T_1) &= \frac{25}{100} \\ p(T_2) &= \frac{40}{100} \\ p(T_3) &= \frac{35}{100} \end{aligned}$$

$$\begin{aligned} p(C_1) &= \frac{50}{100} \\ p(C_2) &= \frac{25}{100} \\ p(C_3) &= \frac{25}{100} \end{aligned}$$

C \ T	T ₁	T ₂	T ₃	Sum n _i
C ₁	0	20	30	50
C ₂	0	20	5	25
C ₃	25	0	0	25
m _j	25	40	35	100

Entropy-Based Measures (I): Conditional Entropy

- **Conditional Entropy:** The cluster-specific entropy, namely the conditional entropy of T with respect to cluster C_i :

$$H(T|C_i) = - \sum_{j=1}^k \left(\frac{n_{ij}}{n_i} \right) \log \left(\frac{n_{ij}}{n_i} \right)$$

How ground truth is distributed within each cluster

n_{ij} ← Ground truth (T)
Cluster (C)

n_{ij}

$$P(T|C_i) = \frac{P(T, C_i)}{P(C_i)} = \frac{n_{ij}/n}{n_i/n} = \frac{n_{ij}}{n_i}$$

C \ T	T ₁	T ₂	T ₃	Sum <u>n_i</u>
C ₁	0	20	30	50
C ₂	0	20	5	25
C ₃	25	0	0	25
m _j	25	40	35	100

Entropy-Based Measures (I): Conditional Entropy

- The conditional entropy of $\mathcal{T}$ given clustering $\mathcal{C}$ is defined as the weighted sum:

$H(\mathcal{T}|\mathcal{C}_i)$

best value = 0

avg. conditional entropy

$$H(\mathcal{T}|\mathcal{C}) = \sum_{i=1}^r \left(\frac{n_i}{n} \right) H(\mathcal{T}|\mathcal{C}_i) = - \sum_{i=1}^r \sum_{j=1}^k p_{ij} \log \left(\frac{p_{ij}}{p_{C_i}} \right)$$

$\frac{n_{ij}}{n}$

$\frac{n_i}{n}$

$$= \underline{H(\mathcal{C}, \mathcal{T}) - H(\mathcal{C})}$$

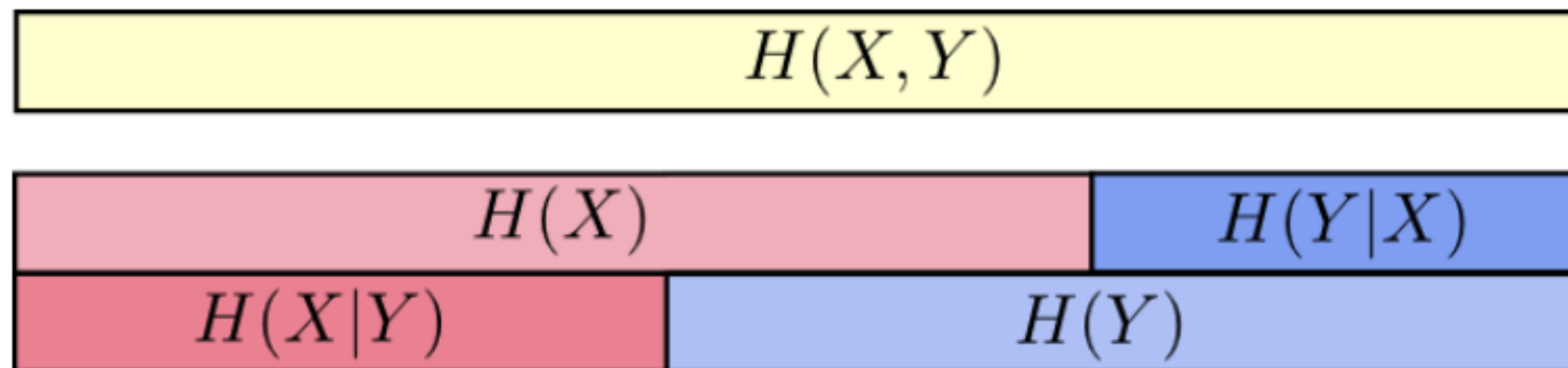
The more clusters members are split into different partitions, the higher the conditional entropy (not a desirable condition and the max value is $\log k$)

$H(\mathcal{T}|\mathcal{C}) = 0$ if and only if $\mathcal{T}$ is completely determined by $\mathcal{C}$, corresponding to the ideal clustering. If $\mathcal{C}$ and $\mathcal{T}$ are independent of each other, then $H(\mathcal{T}|\mathcal{C}) = H(\mathcal{T})$.

$$H(Y|X) = \sum_{x \in X} p(x) H(Y|X = x)$$

$$H(Y|X) = H(X, Y) - H(X)$$

$$\begin{aligned}
 H(\mathcal{T}|\mathcal{C}) &= - \sum_{i=1}^r \sum_{j=1}^k p_{ij} \log \frac{p_{ij}}{p_{c_i}} \\
 &= - \sum_{i=1}^r \sum_{j=1}^k p_{ij} (\log p_{ij} - \log p_{c_i}) = - \sum_{i=1}^r \sum_{j=1}^k p_{ij} (\log p_{ij}) + \sum_{i=1}^r (\log p_{c_i} \sum_{j=1}^k p_{ij}) \\
 &\quad - \sum_{i=1}^r \sum_{j=1}^k p_{ij} \log p_{ij} + \sum_{i=1}^r (p_{c_i} \log p_{c_i}) = H(\mathcal{T}, \mathcal{C}) - H(\mathcal{C})
 \end{aligned}$$



Entropy-Based Measures (I): Mutual Information

$$I(\mathcal{C}, \mathcal{T}) = \frac{H(\mathcal{T})}{H(\mathcal{C})} - \frac{H(\mathcal{T}|\mathcal{C})}{H(\mathcal{C}|\mathcal{T})}$$

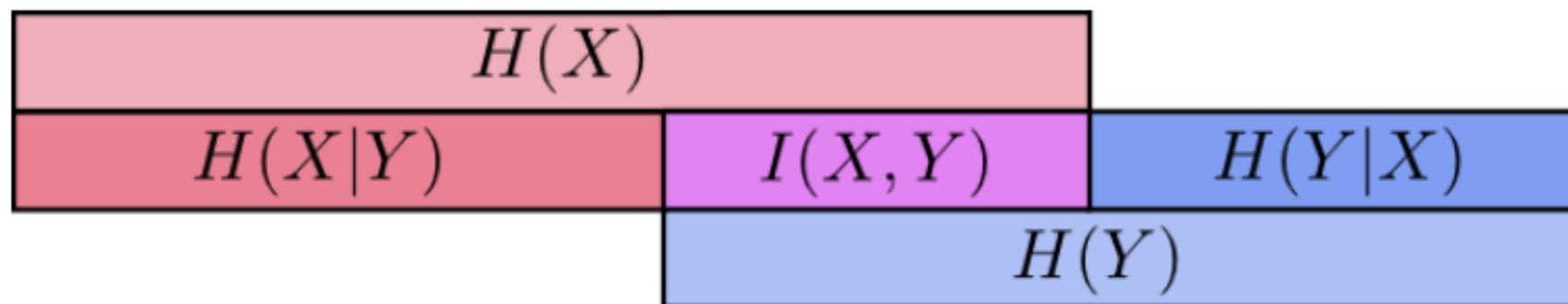
The *mutual information* tries to quantify the amount of shared information between the clustering $\mathcal{C}$ and partitioning $\mathcal{T}$, and it is defined as

as high as possible $\rightarrow$

$$I(\mathcal{C}, \mathcal{T}) = \sum_{i=1}^r \sum_{j=1}^k p_{ij} \log \left(\frac{p_{ij}}{p_{C_i} \cdot p_{T_j}} \right) = \underline{H(\mathcal{T})} - \underline{H(\mathcal{T}|\mathcal{C})}$$

When $\mathcal{C}$ and $\mathcal{T}$ are independent then $p_{ij} = p_{C_i} \cdot p_{T_j}$, and thus $I(\mathcal{C}, \mathcal{T}) = 0$. However, there is no upper bound on the mutual information.

We should do something about this



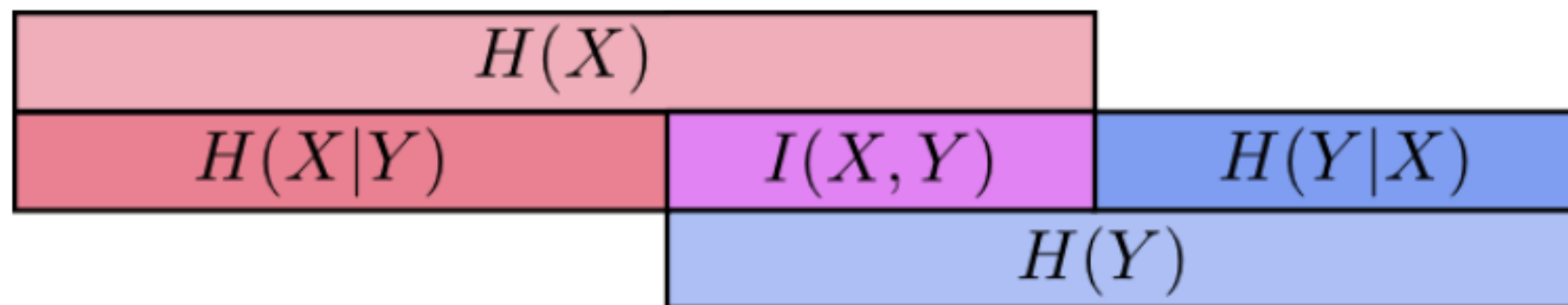
We measure the dependency between the observed joint probability p_{ij} of $\mathcal{C}$ and $\mathcal{T}$, and the expected joint probability $p_{ci} \cdot p_{Tj}$ under the independence assumption

Entropy-Based Measures (I): Mutual Information

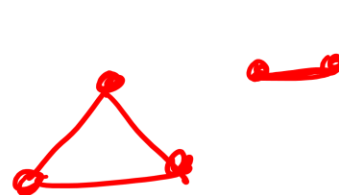
The *normalized mutual information* (NMI) is defined as the geometric mean:

$$\text{NMI}(\mathcal{C}, \mathcal{T}) = \sqrt{\frac{I(\mathcal{C}, \mathcal{T})}{H(\mathcal{C})} \cdot \frac{I(\mathcal{C}, \mathcal{T})}{H(\mathcal{T})}} = \frac{I(\mathcal{C}, \mathcal{T})}{\sqrt{H(\mathcal{C}) \cdot H(\mathcal{T})}}$$

The NMI value lies in the range $[0, 1]$. Values close to 1 indicate a good clustering.



Pairwise Measures



How many edges can we have with n datapoints $\rightarrow \binom{n}{2}$

These metrics evaluate clustering based on how well it groups **pairs of points** that should (or should not) be together.

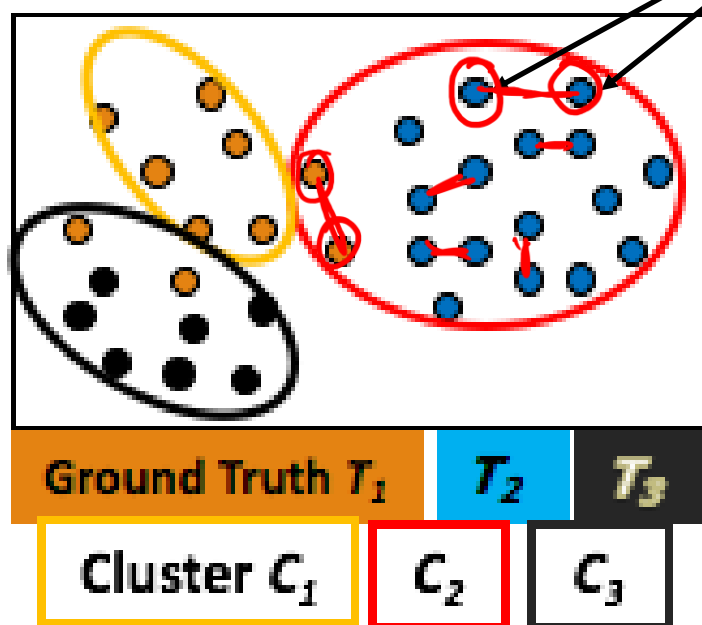
Given clustering $\mathcal{C}$ and ground-truth partitioning $\mathcal{T}$, let $\mathbf{x}_i, \mathbf{x}_j \in \mathbf{D}$ be any two points, with $i \neq j$. Let y_i denote the true partition label and let $\hat{y}_i$ denote the cluster label for point $\mathbf{x}_i$.

True Positives: $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to the same partition in $\mathcal{T}$, and they are also in the same cluster in $\mathcal{C}$. The number of true positive pairs is given as

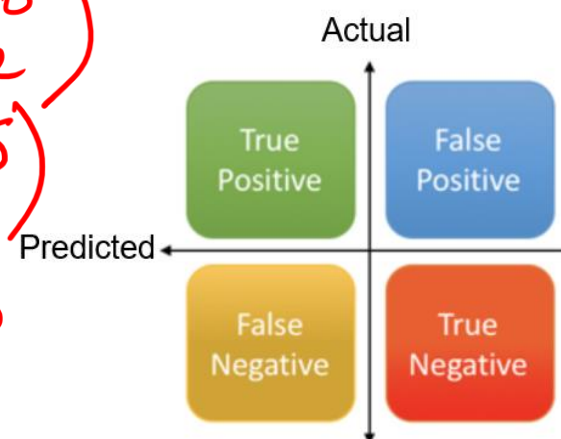
$$TP = |\{(\mathbf{x}_i, \mathbf{x}_j) : y_i = y_j \text{ and } \hat{y}_i = \hat{y}_j\}|$$

Same partition

Same cluster



$$\begin{aligned} & \binom{0}{2} + \binom{20}{2} + \binom{30}{2} \\ & + \binom{0}{2} + \binom{20}{2} + \binom{5}{2} \\ & + \binom{25}{2} + 0 + 0 \end{aligned}$$



n_{ij}

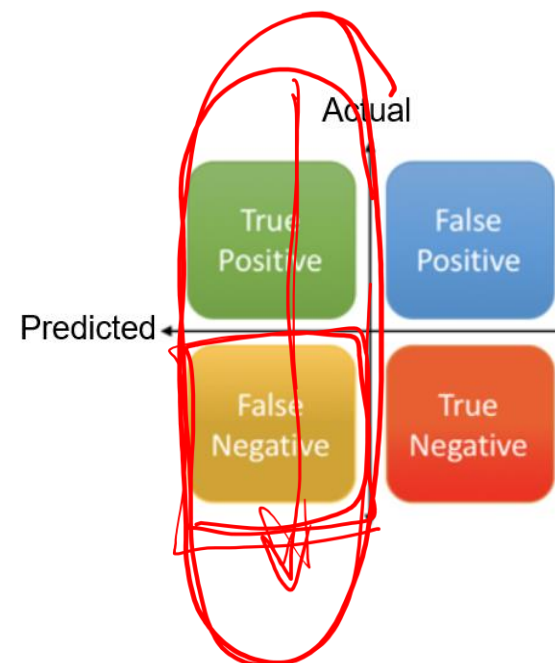
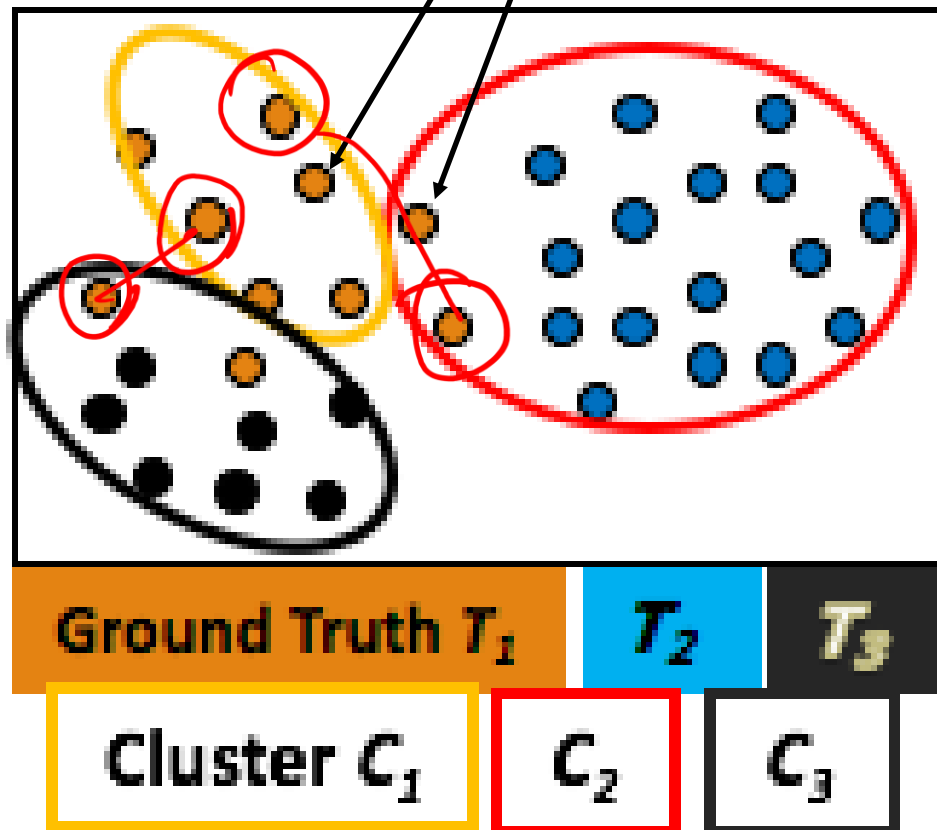
$C \setminus T$	T_1	T_2	T_3	Sum
C_1	0	20	30	50
C_2	0	20	5	25
C_3	25	0	0	25
m_j	25	40	35	100

False Negatives: $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to the same partition in $\mathcal{T}$, but they do not belong to the same cluster in $\mathcal{C}$. The number of all false negative pairs is given as

$$FN = |\{(\mathbf{x}_i, \mathbf{x}_j) : y_i = y_j \text{ and } \hat{y}_i \neq \hat{y}_j\}|$$

Same partition Different cluster

$$\text{TP} + \text{FN} = \binom{25}{2} + \binom{40}{2} + \binom{35}{2}$$

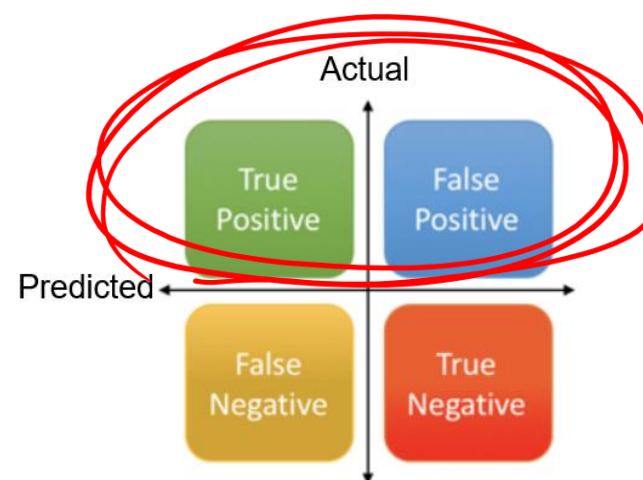
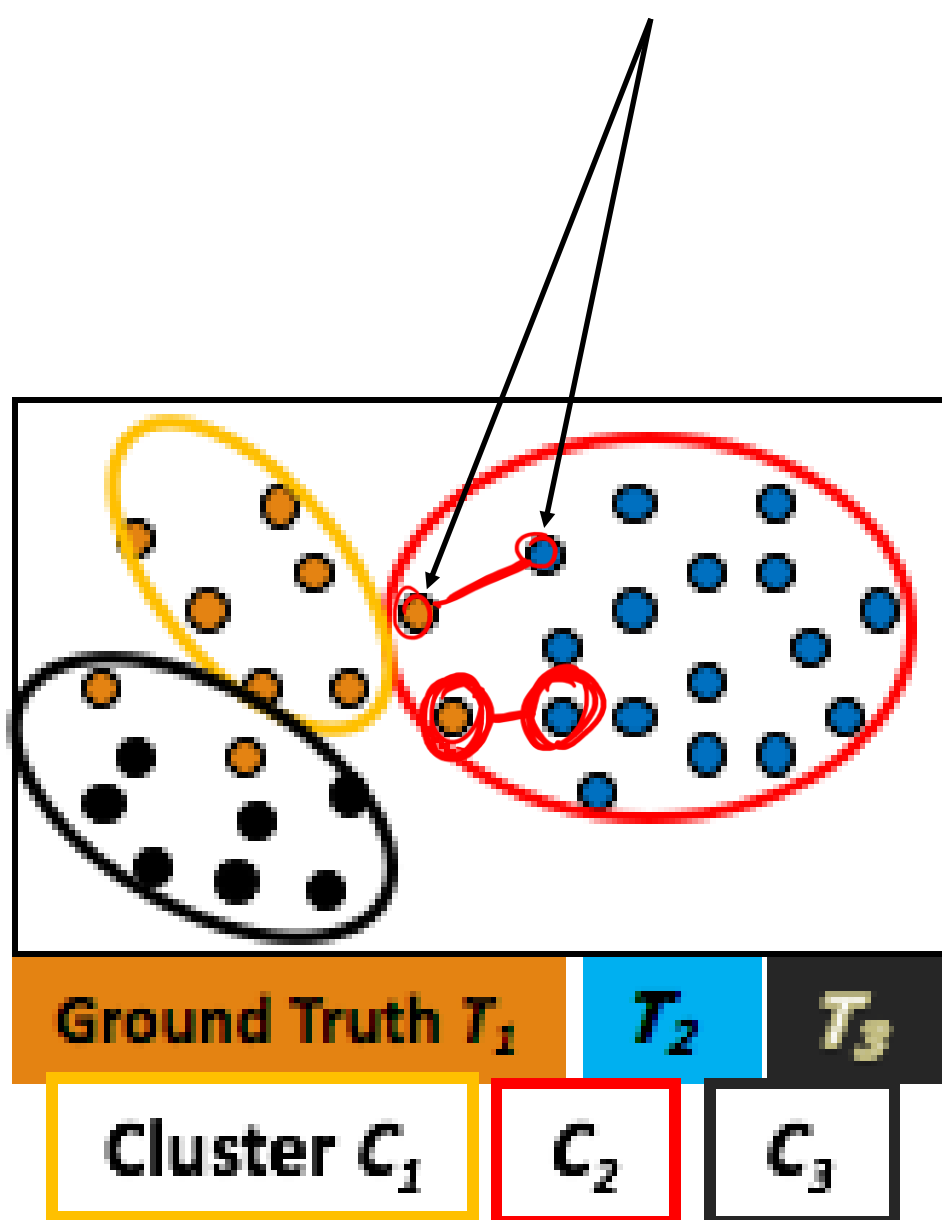


$C \setminus T$	T_1	T_2	T_3	Sum
C_1	0	20	30	50
C_2	0	20	5	25
C_3	25	0	0	25
m_j	25	40	35	100

False Positives: $\mathbf{x}_i$ and $\mathbf{x}_j$ do not belong to the same partition in $\mathcal{T}$, but they do belong to the same cluster in $\mathcal{C}$. The number of false positive pairs is given as

$$FP = |\{(\mathbf{x}_i, \mathbf{x}_j) : \underbrace{y_i \neq y_j}_{\text{Different partition}} \text{ and } \underbrace{\hat{y}_i = \hat{y}_j}_{\text{Same cluster}}\}|$$

$$TP + FP = \binom{50}{2} + \binom{25}{2} + \binom{25}{2}$$

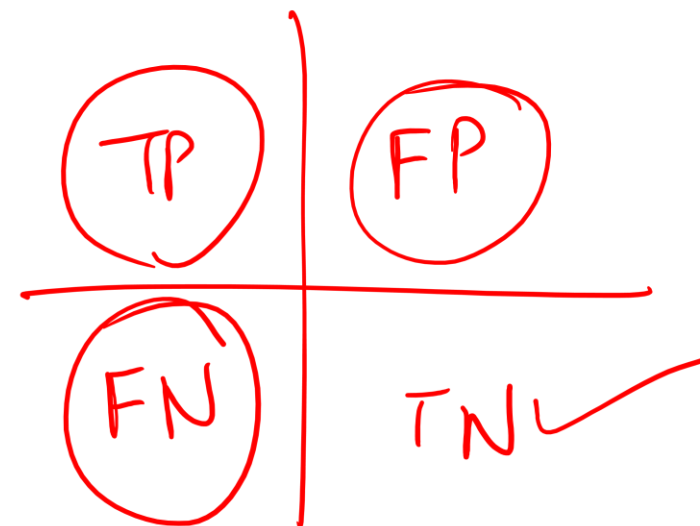
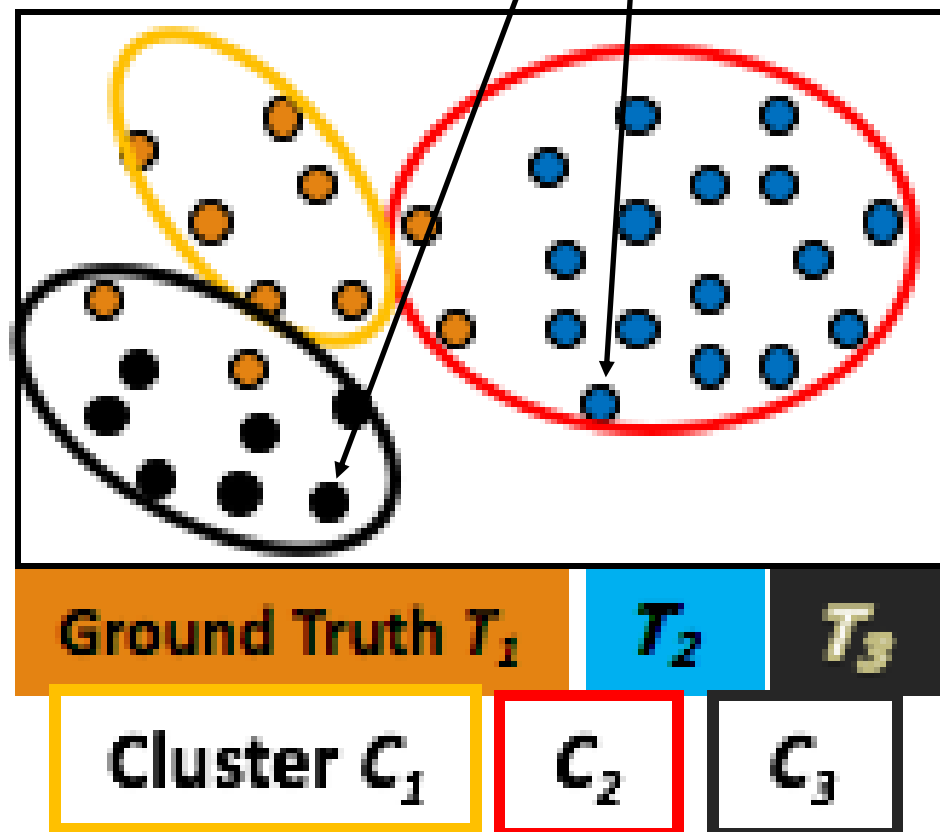


$C \backslash T$	T_1	T_2	T_3	Sum
C_1	0	20	30	50
C_2	0	20	5	25
C_3	25	0	0	25
m_j	25	40	35	100

True Negatives: $\mathbf{x}_i$ and $\mathbf{x}_j$ neither belong to the same partition in $\mathcal{T}$, nor do they belong to the same cluster in $\mathcal{C}$. The number of such true negative pairs is given as

$$TN = |\{(\mathbf{x}_i, \mathbf{x}_j) : \underbrace{y_i \neq y_j}_{\text{Different partition}} \text{ and } \underbrace{\hat{y}_i \neq \hat{y}_j}_{\text{Different cluster}}\}|$$

$$TP + FP + FN + \underline{\underline{TN}} = \binom{100}{2}$$



Pairwise Measures

Because there are $N = \binom{n}{2} = \frac{n(n-1)}{2}$ pairs of points, we have the following identity:

$$N = TP + FN + FP + TN$$

$$TP = \sum_{i=1}^r \sum_{j=1}^k \binom{n_{ij}}{2} = \frac{1}{2} \left(\sum_{i=1}^r \sum_{j=1}^k (n_{ij}^2 - n_{ij}) \right) = \frac{1}{2} \left(\left(\sum_{i=1}^r \sum_{j=1}^k n_{ij}^2 \right) - n \right) \quad TN = N - (TP + FN + FP)$$

$$FN = \sum_{j=1}^k \binom{m_j}{2} - TP$$

$$FP = \sum_{i=1}^r \binom{n_i}{2} - TP$$

$$TN = N - (TP + FN + FP)$$

$C \backslash T$	T_1	T_2	T_3	Sum
C_1	0	20	30	50
C_2	0	20	5	25
C_3	25	0	0	25
m_j	25	40	35	100

$n_{12} = 20$ Points which have same Cluster one and same Partition two

Pairwise Measures: Jaccard Coefficient

Jaccard Coefficient: measures the fraction of true positive point pairs, but after ignoring the true negative:

$$Jaccard = \frac{TP}{TP + FN + FP}$$

(Handwritten red annotations: TP is circled, and TN is circled in the text above)

Measures **similarity** between clustering and ground truth — focuses on positive pairs only (ignores TN).

A **perfect clustering** gives Jaccard = 1.

It answers: “Of all pairs that should or were grouped together, what fraction are correctly grouped?”

Pairwise Measures: Rand

Rand Statistic: measures the fraction of true positives and true negatives over all point pairs:

$$Rand = \frac{TP + TN}{N}$$

no. of pairs of datapoints
in diff. cluster & position
could be high

- Measures the **fraction of agreements** (both same-cluster and different-cluster pairs) between clustering and ground truth.
- Includes true negatives (pairs correctly not grouped together).
- Conceptually similar to **accuracy** in classification.
- **Perfect clustering** → Rand = 1.
- **Downside:** Because TN pairs dominate (most pairs are in different clusters), Rand Index can appear high even if clustering is mediocre.

Pairwise Measures: FM

Fowlkes-Mallows Measure: Define the overall *pairwise precision* and *pairwise recall* values for a clustering $\mathcal{C}$, as follows:

$$prec = TP / TP + FP$$

$$recall = TP / TP + FN$$

The Fowlkes–Mallows (FM) measure is defined as the geometric mean of the pairwise precision and recall

$$FM = \sqrt{prec \cdot recall} = \frac{TP}{\sqrt{(TP + FN)(TP + FP)}}$$

- It combines **pairwise precision** and **pairwise recall** — just like the F1 score did for classification.
- A **perfect clustering** gives $FM = 1$.

Jaccard and **FM** are preferred when **number of TNs is large**, since they avoid the dominance of negatives. **Rand Index** is simple and interpretable but can be inflated when most pairs are different.

Summary

- External measures for clustering evaluation
 - Matching-based measures
 - Entropy-based measures
 - Pairwise measures
- Internal measures for clustering evaluation
 - Graph-based measures
 - Davies-Bouldin Index
 - Silhouette Coefficient

0 pre 90% recall →

Quick Knowledge Check

1. **What does purity measure?** A. Compactness of clusters. B. Homogeneity with respect to ground truth. C. Separation between clusters
2. **What problem does maximum ^{purity} matching fix?** A. Duplicate cluster-partition matches. B. Small cluster bias. C. Missing cluster labels
3. **Why is the harmonic mean used in F-measure?** A. To emphasize the larger value. B. To balance precision and recall fairly. C. To simplify computation
4. **What does conditional entropy $H(T | C)$ represent?** A. Uncertainty of true labels within clusters. B. Compactness of clusters. C. Distance between centroids
5. **Higher conditional entropy implies:** A. Better clustering. B. Poorer clustering. C. No change in clustering quality
6. **Why doesn't mutual information have a universal upper bound in clustering evaluation?** A. It depends on the number and size of clusters, which can vary. B. It ignores entropy in computation
7. **Jaccard coefficient ignores:** A. True negatives. B. False positives. C. False negatives
8. **Rand Index is most similar to:** A. Accuracy. B. Precision. C. Recall
9. **Why can Rand Index appear high even for poor clustering?** A. TN pairs dominate. B. FP pairs dominate. C. FN pairs dominate